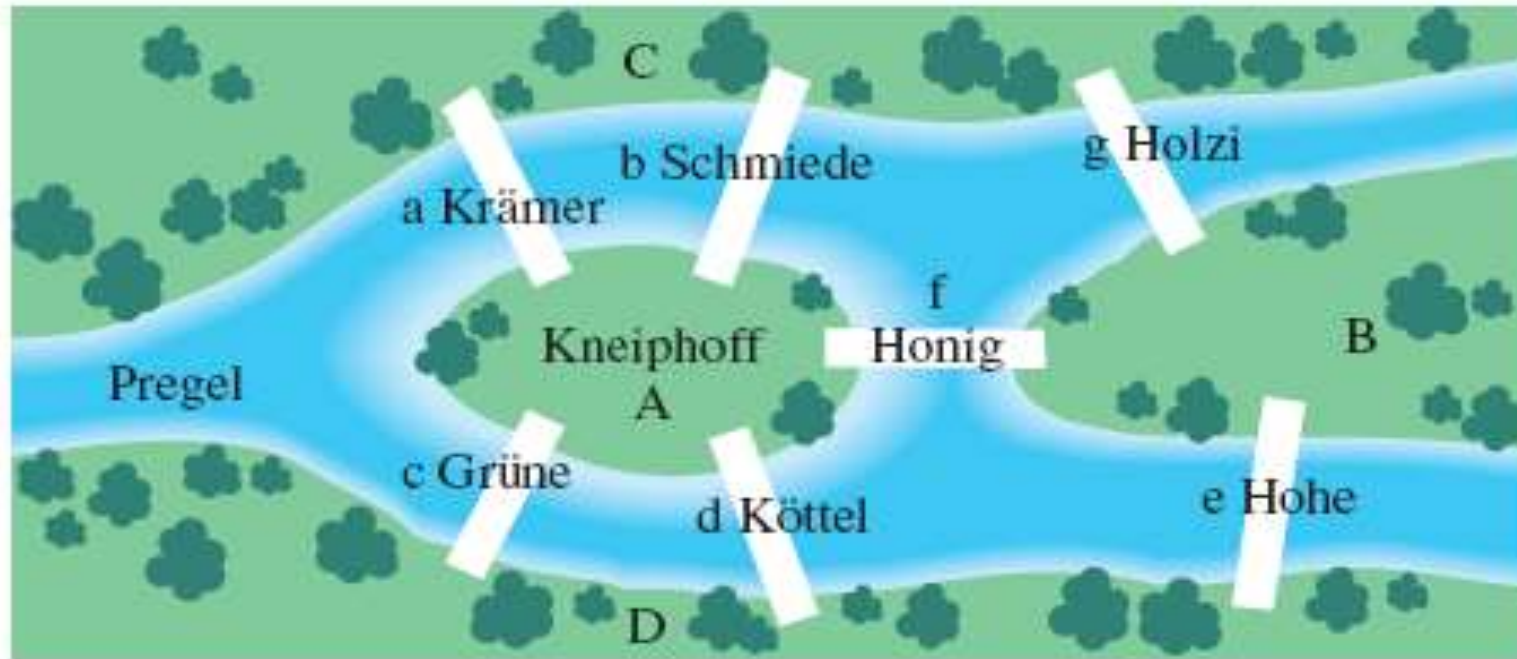


Euler and Hamilton Paths

Konigsberg Bridge Problem

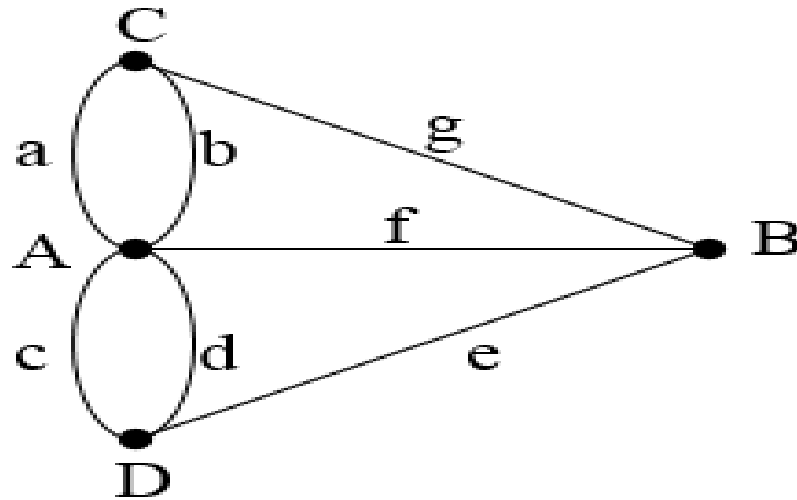
- In the early 1700s, the city of Konigsberg was the capital of East Prussia. The river Pregel ran through the city in two branches with an island between the branches (see figure on next slide).
- There were seven bridges joining various parts of the city. The following problem was well known. Is it possible for a citizen of Konigsberg to take a stroll through the city, crossing each bridge exactly once, and beginning and ending at the same place?

Map of Königsberg



Graph of the Map

- The vertices represent landmasses. The edges represent the bridges



Euler Path and Euler Circuit

- An **Euler path** in a graph is a path that uses every edge of the graph exactly once.
- An **Euler circuit** in a graph is a circuit that uses every edge of the graph exactly once.

Term	Initial and terminal vertices the same	Must include every edge	Repeated vertices allowed
Eulerian Circuit	Yes	Yes	Yes
Eulerian Trail	No	Yes	Yes

Hamilton paths

A path $x_0, x_1, \dots, x_{n-1}, x_n$ in the graph $G = (V, E)$ is called Hamilton path if $V = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ and $x_i \neq x_j$ for $0 \leq i < j \leq n$. A circuit $x_0, x_1, \dots, x_{n-1}, x_n, x_0$, with $n > 1$, in a graph $G = (V, E)$ is a Hamilton circuit if $x_0, x_1, \dots, x_{n-1}, x_n$ is a Hamilton path.

A graph with a vertex of degree one cannot have a Hamilton circuit.

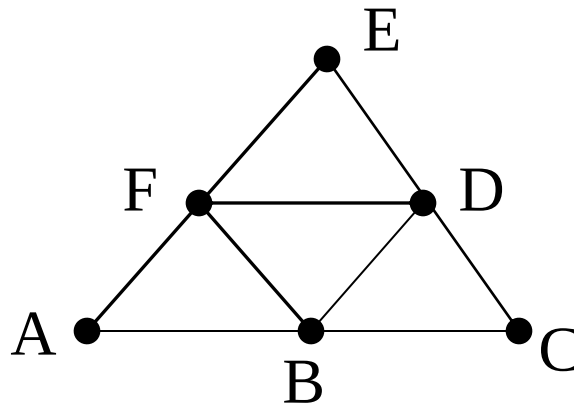
If a vertex in a graph has degree 2, then both edges incident with this vertex must be part of Hamilton cycle.

Term	Initial and terminal vertices the same	Must include every edge	Repeated vertices allowed
Eulerian Circuit	Yes	Yes	Yes
Hamiltonian Cycle	YES	NO	NO

Example: Recognizing Euler Circuits

Consider the graph below.

- a) Is $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow A$ an Euler circuit?
- b) Does the graph have an Euler circuit?



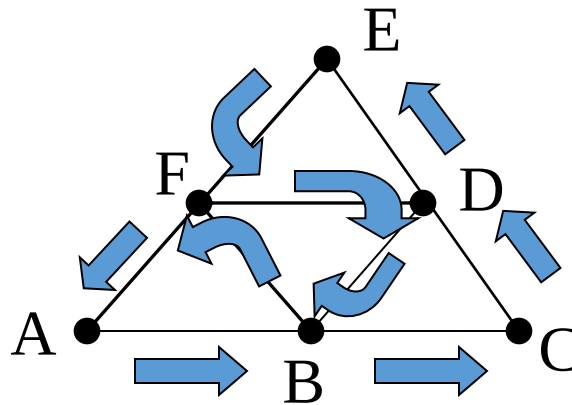
Example: Recognizing Euler Circuits

Solution

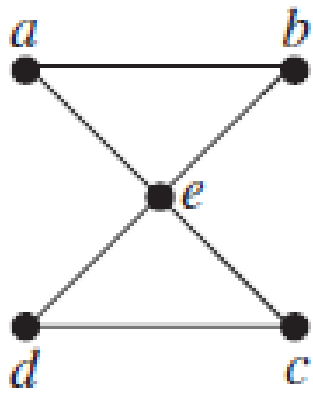
a) No, it does not use edge BD.

b) Yes, the circuit

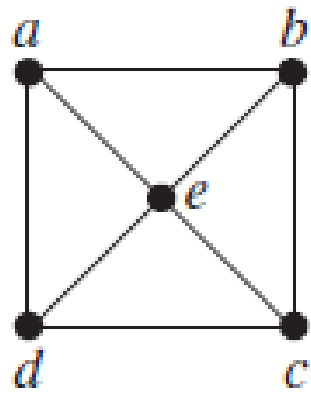
$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow D \rightarrow B \rightarrow F \rightarrow A$
is an Euler circuit.



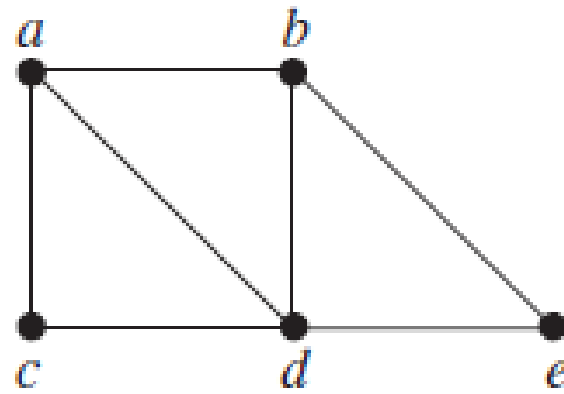
Which of the undirected graphs in Figure have an Euler circuit? Of those that do not, which have an Euler path?



G_1



G_2



G_3

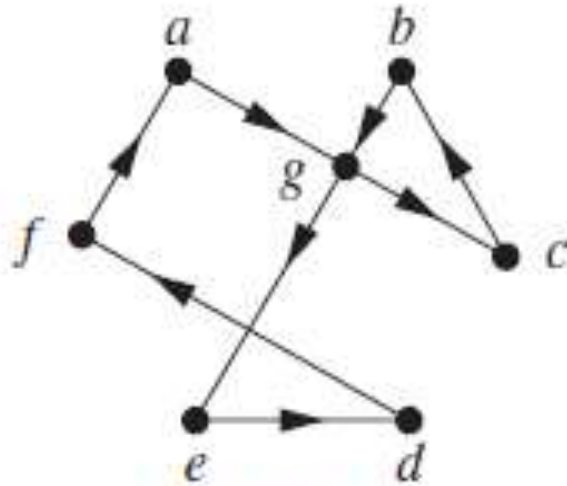
Solution:

- The graph G_1 has an Euler circuit, for example, a, e, c, d, e, b, a . Neither of the graphs G_2 or G_3 has an Euler circuit.
- However, G_3 has an Euler path, namely, a, c, d, e, b, d, a, b .
- G_2 does not have an Euler path.

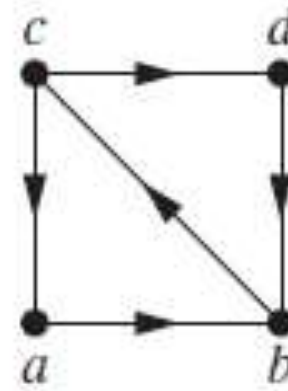
Which of the directed graphs in Figure have an Euler circuit? Of those that do not, which have an Euler path?



H_1



H_2



H_3

Solution:

- The graph H_2 has an Euler circuit, for example, $a, g, c, b, g, e, d, f, a$. Neither H_1 nor H_3 has an Euler circuit.
- H_3 has an Euler path, namely, c, a, b, c, d, b , but H_1 does not have Euler path.

NECESSARY AND SUFFICIENT CONDITION FOR EULER CIRCUITS AND PATHS

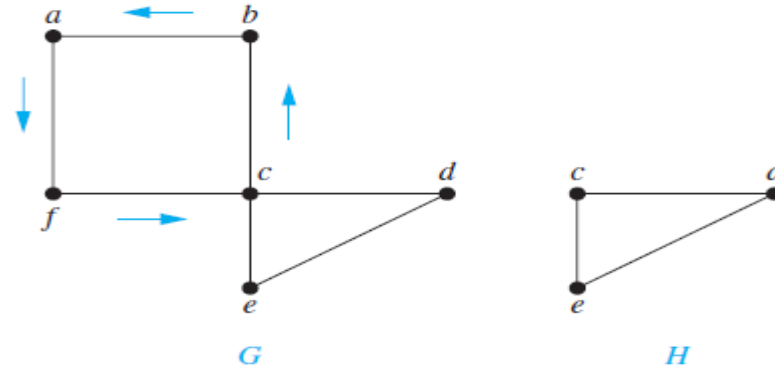
A connected multigraph has and Euler circuit if and only if each of its vertices has even degree.

Proof:

- Take a connected multigraph $G=(V,E)$ where V and E are finite. We can prove a theorem in two parts.
- First we prove that if a connected multigraph has an Euler circuit, then all the vertices have even degree.
- For this, Euler circuit begins with a vertex a and continues with an edge incident with a , say $\{a,b\}$.
- The edge $\{a,b\}$ contributes $\deg(a)$ is one.
- Each time the circuit passes through a vertex, it contributes the degree of vertex of two, because the circuit enters via an edge incident with this vertex and leaves via another such edge.
- Finally, the circuit terminates where it started, contributing degree of vertex equal to one.
- Therefore $\deg(a)$ must be even. Because the circuit contributes one when it begins and one when it ends.
- A vertex other than a has even degree because the circuit contributes degree of vertex equal to two when it passes through the vertex.
- We conclude that if a connected graph has an Euler circuit, then every vertex must have even degree.

- Now we try to prove that if all the vertices in connected multigraph have even degree, there exist Euler circuit.
- For this take a multi graph G with all the vertices having even degree.
- To make a circuit start at arbitrary vertex say a of G .
- Now start from the vertex $a=x_0$ and arbitrarily choose other vertex x_1 to form an edge $\{x_0, x_1\}$. Continue building the simple path $\{x_0, x_1\}$, $\{x_1, x_2\}$, $\dots\dots\{x_{n-1}, x_n\}$. This path terminates since it has a finite number of edges.
- It begins at a with an edge $\{a,x\}$ and terminates at a with some edge $\{y,a\}$. This is correct since every vertex has even degree in the graph we are considering. If an edge left some vertex then there must be an edge entering that vertex to make the degree even.
- If the circuit has all the edges of the graph G in it, then simple circuit is itself an Euler circuit.
- If all the edges are not in the circuit, then we have next possibility.

- Now, consider the subgraph, say H that is formed by removing all the edges that are already in the simple circuit formed above and by removing the isolated vertices after edges are removed.

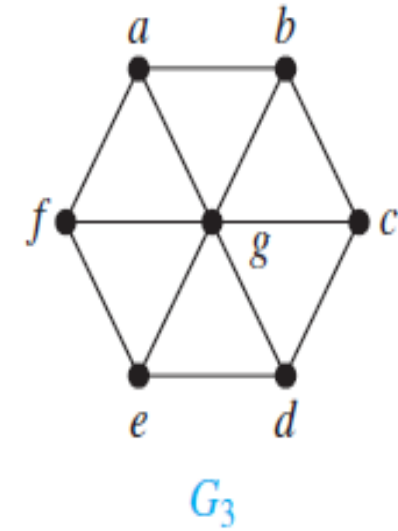
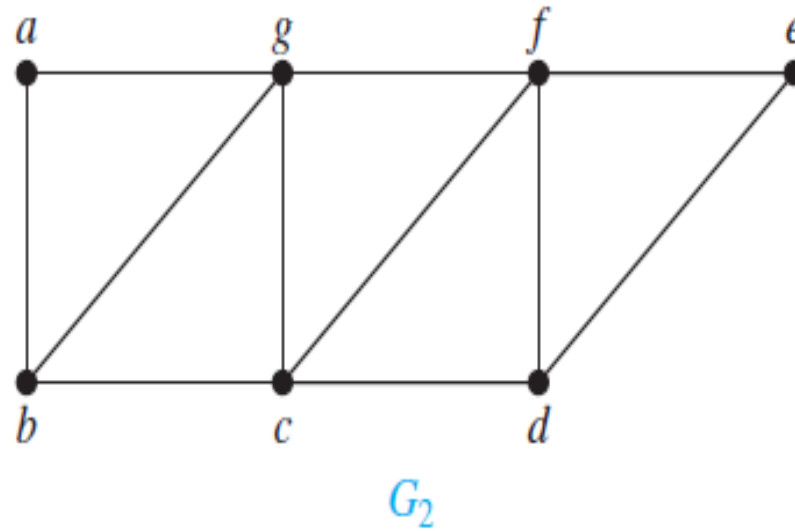
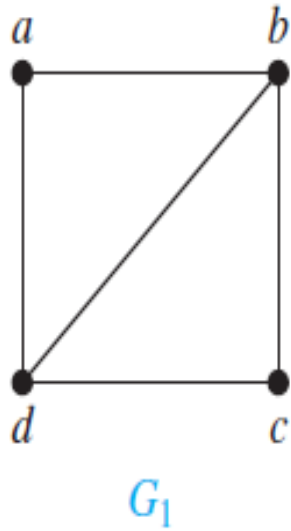


- When we delete the circuit a, f, c, b, a from the graph, we obtain a subgraph.
- Since it is subgraph of original graph G is connected, there must be at least one vertex of H that is common with the circuit we have formed.
- Let w be such a vertex (in this example, $w=c$). Every vertex in H has even degree since it is a subgraph of original graph.
- Beginning at w , construct a simple path in H by choosing edges as long as possible, as was done in G . This path must terminate at w .
- c, d, e, c is a path in H . A circuit is formed in G by splicing the circuit in H with the original circuit in G .
- Continue the process until all edges have been used which produces Euler circuit.
- Hence if every vertices of a connected graph has an even degree then it has an Euler circuit. This concludes the proof.

- **A connected multigraph has an Euler path but not Euler circuit if and only if it has exactly two vertices of odd degree.**
- **Proof:** This fact can be proved if we can prove that first, if the connected multigraph has Euler path exactly two vertices have odd degree and second if the connected multigraph has exactly two vertices of odd degree, then it has Euler path.
- Now, if the graph (in this proof graph means connected multigraph) has an Euler path say from **a** to **z** but not Euler circuit, then it must pass through every edge exactly once.
- In this scenario, the first edge in the path contributes one to the degree of vertex **a**, and at all other time when other edges pass through vertex **a** it contributes twice to the degree of **a**, hence we can say that degree of **a** is odd.
- Similarly, the last edge in the path coming to **z** contributes one to the degree of **z**, all the other edges contributes two, one for entering and one for leaving.
- Here, also the degree of last vertex is odd. All the other vertices other than **a** and **z** must have even degree since the edges in this vertices enter and leave the vertex contributing two to the degree every time the vertices are made.
- Hence, if there is any Euler path but not an Euler circuit, exactly two vertices of the graph have odd degree.

- Secondly, if exactly two vertices of a graph have odd degree and let's consider they are a and z .
- Now, consider another graph that adds an edge $\{a, z\}$ to the original path, then the newly formed graph will have every vertex of even degree.
- So there exists Euler circuit in the new graph and the removal of new edge gives us the Euler path in the original graph.
- Hence, if exactly two vertices of the graph have an odd degree, then the graph has an Euler path but not Euler circuit.
- This includes the proof.

Which graphs shown in Figure have an Euler path?



- G_1 contains exactly two vertices of odd degree, namely, b and d . Hence, it has an Euler path that must have b and d as its endpoints. One such Euler path is d, a, b, c, d, b .
- Similarly, G_2 has exactly two vertices of odd degree, namely, b and d . So it has an Euler path that must have b and d as endpoints. One such Euler path is $b, a, g, f, e, d, c, g, b, c, f, d$.
- G_3 has no Euler path because it has six vertices of odd degree.

Hamilton Paths and Circuits

- A *Hamilton path* in a graph G is a path which visits every vertex in G exactly once.
- A *Hamilton circuit* is a Hamilton path that returns to its start.
- A graph with a vertex of degree one cannot have a Hamilton circuit.
- If a vertex in a graph has degree 2 then both the edges incident with this vertex must be part of the Hamiltonian cycle.

Sir William Rowan Hamilton

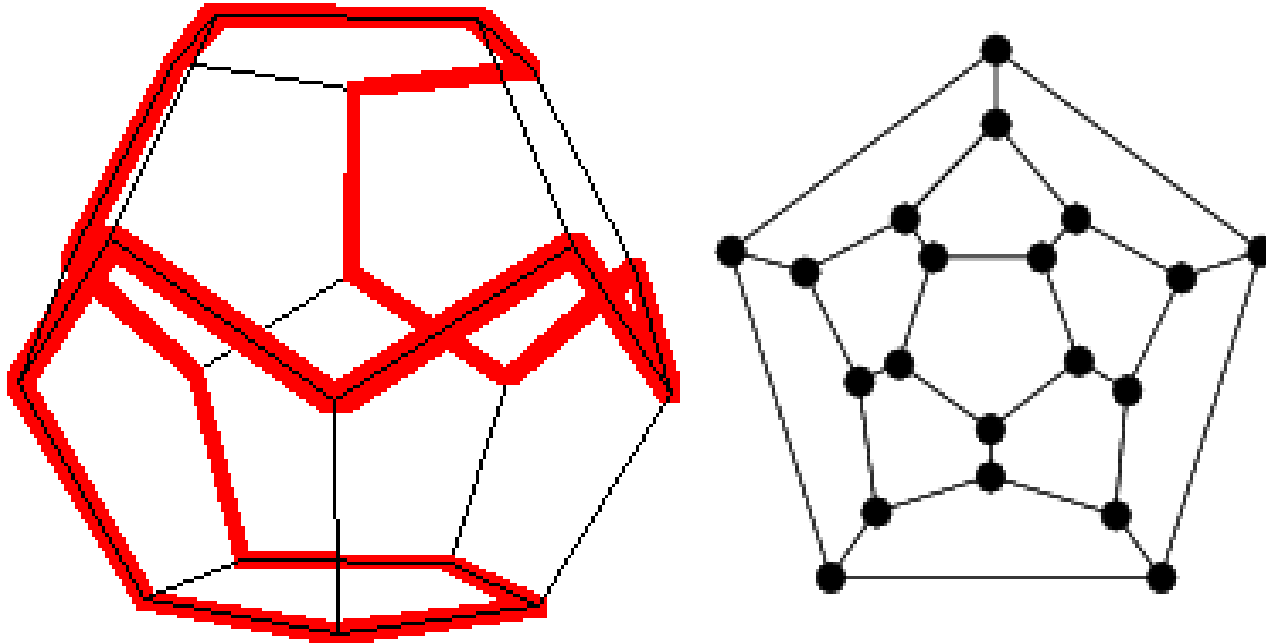
- In the 19th century, an Irishman named Sir William Rowan Hamilton (1805-1865) invented a game called the Icosian game.
- The game consisted of a graph in which the vertices represented in Europe.



The Icosian Game

- The object of the game was to find a path that visited each of the 20 vertices exactly once.
- In honor of Hamilton and the his game, a path that uses each vertex of a graph exactly once is known as a **Hamiltonian path**.
- If the path ends at the starting vertex, it is called a **Hamiltonian circuit**.

Hamilton Circuits

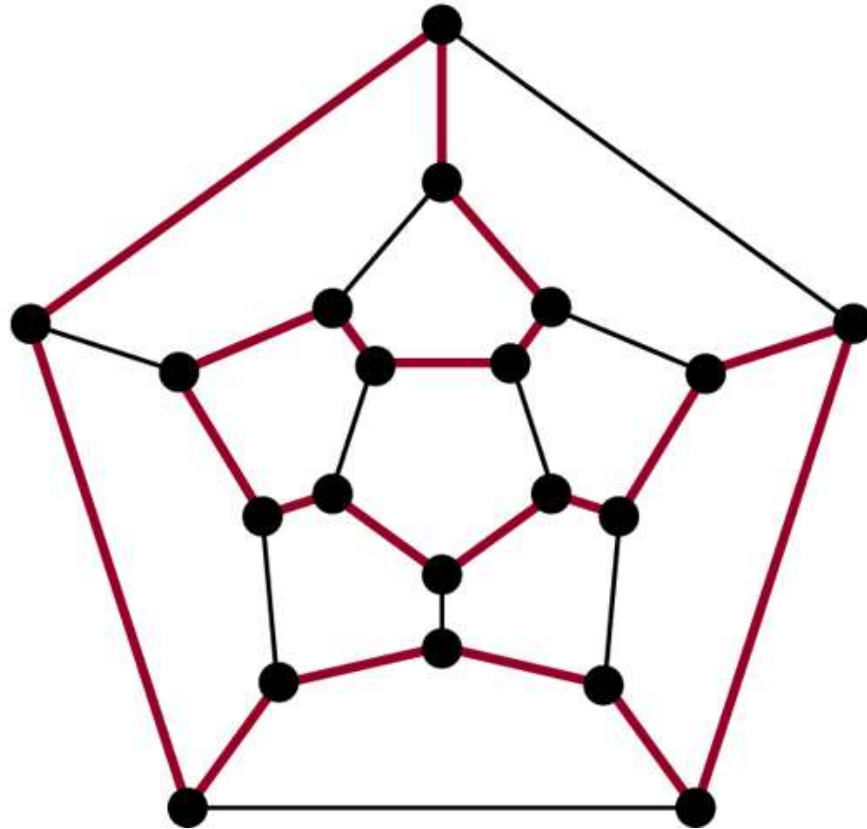


Dodecahedron puzzle and its equivalent graph

Is there a circuit in this graph that passes through each vertex exactly once?

Hamilton Circuits

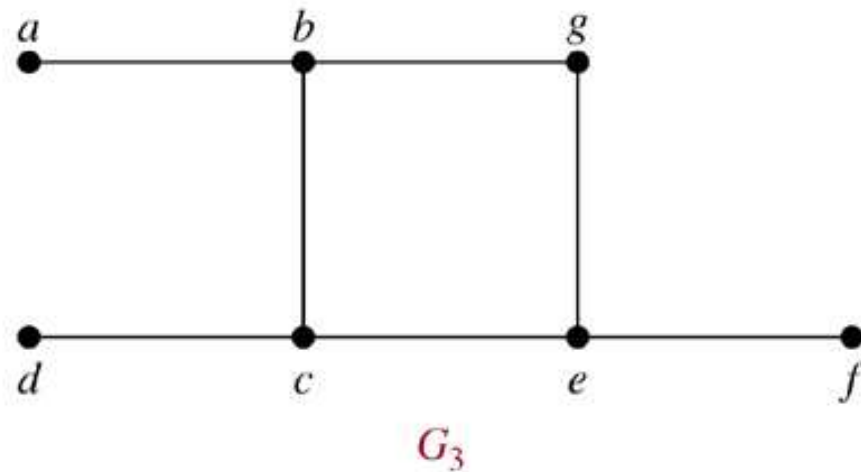
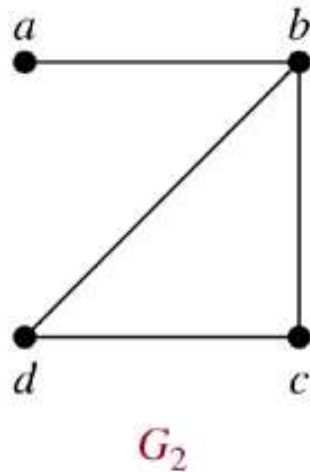
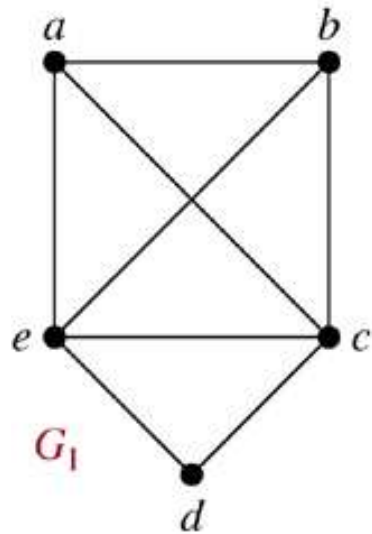
© The McGraw-Hill Companies, Inc. all rights reserved.



Yes; this is a circuit that passes through each vertex exactly once.

Finding Hamilton Circuits

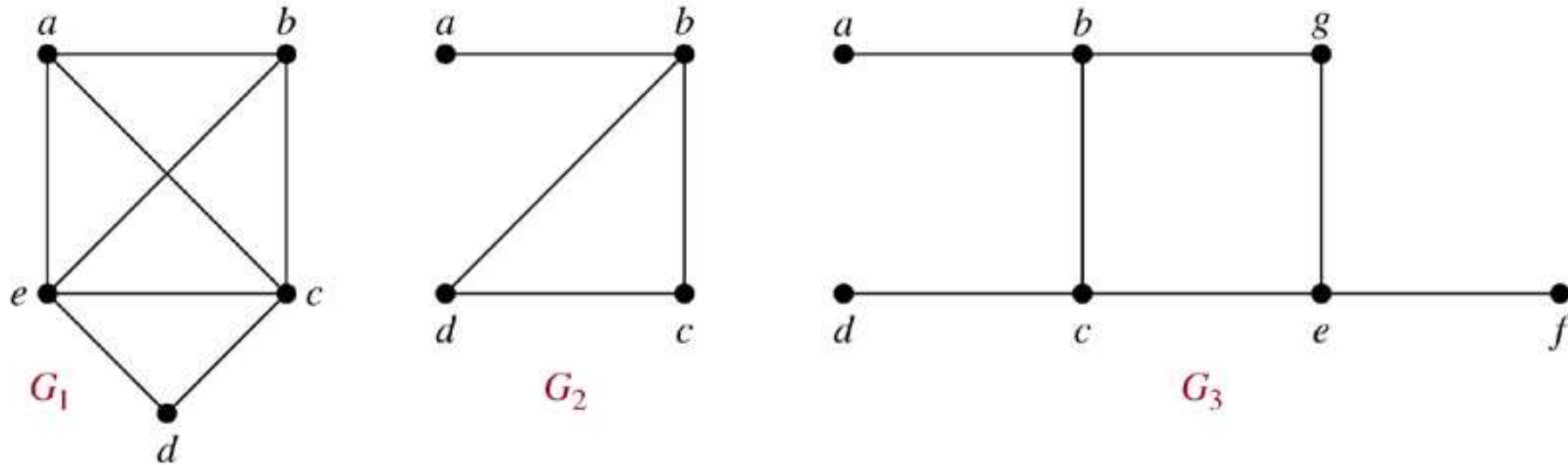
© The McGraw-Hill Companies, Inc. all rights reserved.



Which of these three figures has a Hamilton circuit?
Of, if no Hamilton circuit, a Hamilton path?

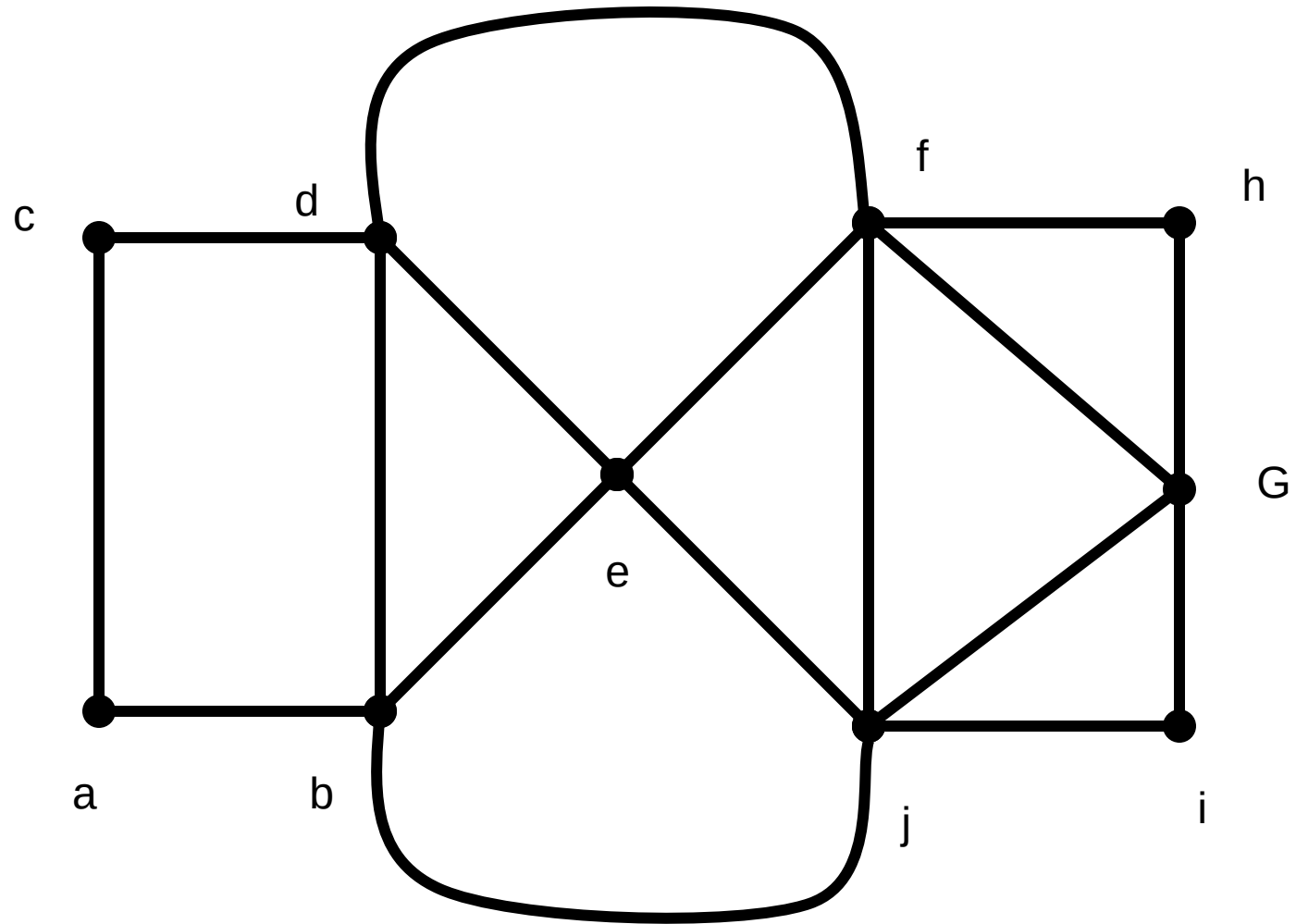
Finding Hamilton Circuits

© The McGraw-Hill Companies, Inc. all rights reserved.

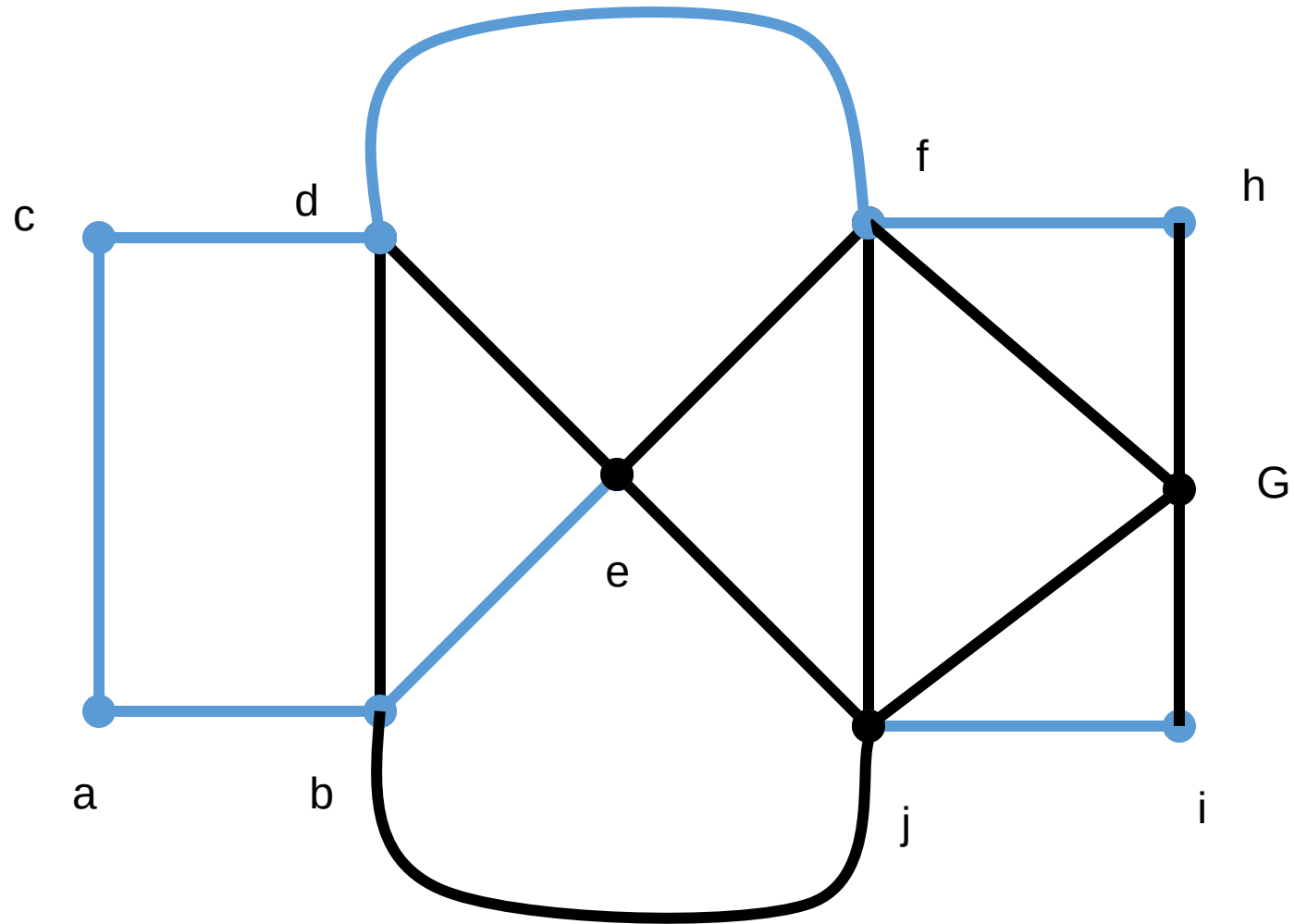


- G_1 has a Hamilton circuit: a, b, c, d, e, a
- G_2 does not have a Hamilton circuit, but does have a Hamilton path: a, b, c, d
- G_3 has neither.

Exploration



Exploration



One Path Works!

- Notice that only one path meets these criteria.
- It is path G, h, f, d, c, a, b, e, j, i, G.
- Also notice, that is not necessary that every edge of the graph be traversed when visiting each vertex exactly once.

Show that K_n has a Hamilton circuit whenever $n \geq 3$

- We can form a Hamilton circuit in K_n beginning at any vertex. Such a circuit can be built by visiting vertices in any order we choose, and the path begins and ends at the same vertex and visits each other vertex exactly once.
- This is possible because there are edges in K_n between any two vertices.

Finding The Hamiltonian

- Since each of the five vertices of the graph has degrees of at least $5/2$, the graph has a Hamiltonian circuit.
- Unfortunately, the theorem does not tell us how to find the circuit.

Hamiltonian Circuits

- If a graph has some vertices with degree less than $n/2$, the theorem does not apply.
- The second two of the figures that are drawn have vertices that have a degree less than $5/2$, so no conclusion can be drawn.
- By inspection, the second figure has a Hamiltonian circuit but the last figure does not.